RAG-Based Customer Support Assistant

(Low level Design)

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Course: Internship Project

Date: April 2026

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Module-Level Design

Document Processing Module

- Load PDF
- Extract text
- Clean data

Chunking Module

- Split text into logical sections
- Preserve semantic meaning

Embedding Module

- Convert chunks to vectors

Vector Storage Module

- Store embeddings in ChromaDB

Retrieval Module

- Perform similarity search

Query Processing Module

- Accept user query
- Preprocess text

Graph Execution Module

- Executes LangGraph nodes

HITL Module

- Escalation handling

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2. Data Structures

Document Representation

```
{
  "doc_id": "pdf_1",
  "content": "raw text"
}
```

Chunk Format

```
{
  "chunk_id": "c1",
  "text": "Refund policy text",
  "metadata": "refund"
}
```

Embedding Structure

```
{
  "vector": [0.12, 0.98, ...]
}
```

Query Schema

```
{
  "query": "What is refund policy?",
  "intent": "refund"
}
```

Graph State Object

```
{
  "query": "",
  "context": [],
  "answer": "",
  "confidence": 0.0,
  "escalate": false
}
```

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3. Workflow Design (LangGraph)

◈ Nodes

- Input Node
- Intent Classification Node
- Retrieval Node
- LLM Node

- HITL Node
- Output Node

◈ Flow

Input → Intent → Retrieval → LLM → Confidence Check → HITL / Output

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◈ State Passing

Each node updates shared state object.

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4. Conditional Routing Logic

✓ Answer Generation Criteria

- High similarity score
- Relevant chunk found

⚠ Escalation Criteria

- Low confidence (< threshold)
- No relevant chunk found
- Ambiguous query

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5. HITL Design

◈ When Escalation Happens

- Model confidence is low
- Query not found in knowledge base

◈ After Escalation

- Query sent to human agent dashboard
- Human provides response

◈ Integration

- Human response replaces LLM output
- Stored for future training

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6. API / Interface Design

Input Format

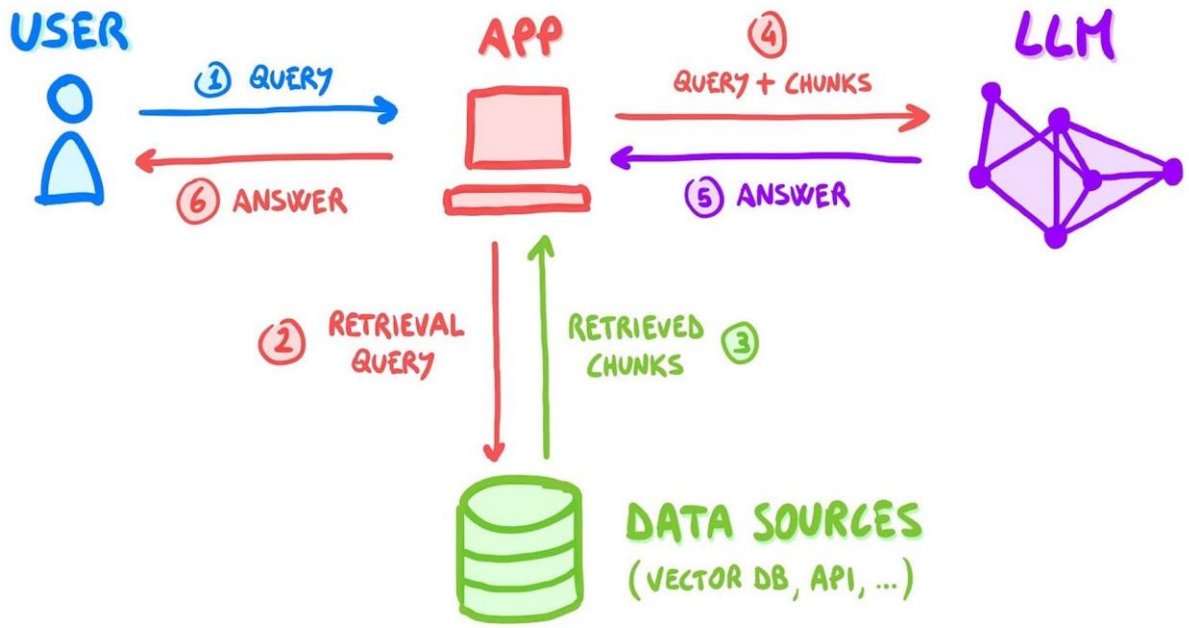
```
{
  "query": "What is refund policy?"
}
```

Output Format

```
{
  "answer": "Customers can request refund within 7 days.",
  "confidence": 0.87,
  "source": "chunk_12"
}
```

Interaction Flow

User → UI → Pipeline → Retrieval → LLM → Response → UI



7. Error Handling

✕ Missing Data

- Return fallback message

✕ No Relevant Chunks

- Trigger HITL or “Not found”

✕ LLM Failure

- Retry mechanism